NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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Abstract. In this manuscript, we derive formulas for multifold sums of powers by utilizing Newton's interpolation formula. Furthermore, we provide formulas for multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers.

Contents

Date: June 27, 2026.

2010 *Mathematics Subject Classification.* 05A19, 05A10, 41A15, 11B83.

Key words and phrases. Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Newton's series, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Combinatorics, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, OEIS.

DOI: https://doi.org/10.5281/zenodo.18040979

1. Auxiliary lemmas

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\begin{aligned}\Sigma^0 n^m &= n^m,\\ \Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,\\ \Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.\end{aligned}$$

We utilize the following notation for rising factorials

Definition 1.2 (Rising factorials)**.** *For integers n, k*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0,\\ 1, & \text{if } k = 0,\\ \prod_{j=0}^{k-1}(n+j), & \text{if } k > 0.\end{cases}$$

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [2].

Lemma 1.3 (Forward Newton's formula)**.** *Let $f\colon \mathbb{Z} \to \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \Delta^k f(t) = \sum_{k=0}^{\infty} \binom{n-t}{k} \Delta^k f(t),$$

where the k-th forward difference $\Delta^k f(t)$ of f evaluated at t is defined by

$$\Delta^k f(t) = \sum_{j=0}^{k} (-1)^{k-j} \tbinom{k}{j} f(t+j).$$

Proof. Originally revealed by Sir Isaac Newton in [3]. Expressed in modern mathematical notation by J. F. Steffensen in [4, section 2, eq. (19)]. By using identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t}{k}$, we get RHS. $\square$

Lemma 1.4 (Forward Newton's formula for powers)**.** *For non-negative integers $n, m,$ and an arbitrary integer t*

$$n^m = \sum_{k=0}^{m} \binom{n-t}{k} \Delta^k t^m.$$

Proof. Special case of Lemma (1.3) for powers. □

Lemma 1.5 (Generalized hockey-stick identity)**.** *For arbitrary integers a, b and j,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. By hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 1.6 (Forward hockey-stick identity)**.** *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n} \binom{j-t+r}{k+r} = \binom{n-t+r+1}{k+r+1} - \binom{1-t+r}{k+r+1}.$$

Proof. By setting $a = 1-t+r$, and $b = n-t+1$ into Lemma (1.5). □

Lemma 1.7 (Multifold sums of zero powers)**.** *For non-negative integers r, n*

$$\Sigma^r \, n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) Let $r = 0$, then we have $\Sigma^0 \, n^0 = 1$, by definition (1.1).

(2) Let $r = 1$, then we have $\Sigma^1 \, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) Let $r = 2$, then we have $\Sigma^2 \, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) Let $r = 3$, then we have $\Sigma^3 \, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) Similarly, for arbitrary integer r, by hockey-stick identity $\sum_{k=1}^{n} \binom{k}{r} = \binom{n+1}{r+1}$, yields $\Sigma^r \, n^0 = \sum_{k=1}^{n} \Sigma^{r-1} \, k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

This completes the proof. □

Lemma 1.8 (Upper Binomial negation)**.** *For integers n, k*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$$

Proof. See [5, p. 164, eq. (5.14)]. □

Convention 1.9. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x, including zero, in Concrete Mathematics [5, p. 162], and in [6].

2. Introduction

3. Main results

We start our discussion from Lemma (1.4), which yields forward Newton's interpolation formula for powers. For example,

$$n^3 = 0\tbinom{n-0}{0} + 1\tbinom{n-0}{1} + 6\tbinom{n-0}{2} + 6\tbinom{n-0}{3},$$

$$n^3 = 1\tbinom{n-1}{0} + 7\tbinom{n-1}{1} + 12\tbinom{n-1}{2} + 6\tbinom{n-1}{3},$$

$$n^3 = 8\tbinom{n-2}{0} + 19\tbinom{n-2}{1} + 18\tbinom{n-2}{2} + 6\tbinom{n-2}{3},$$

$$n^3 = 27\tbinom{n-3}{0} + 37\tbinom{n-3}{1} + 24\tbinom{n-3}{2} + 6\tbinom{n-3}{3}.$$

Thus, the transition to formula for sums of powers is quite straightforward,

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\sum_{j=1}^{n} \binom{j-t}{k} \right] \Delta^k t^m.$$

Therefore, by setting $r = 0$ into Lemma (1.6) yields closed form of ordinary sums of powers

Proposition 3.1 (Ordinary sums of powers)**.** *For non-negative integers n, m, and an arbitrary integer t*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\binom{n-t+1}{k+1} - \binom{1-t}{k+1} \right] \Delta^k t^m.$$

We may see that the basis $\binom{j-t+r}{k+r}$ for repeatedly applied Lemma (1.6) was indeed chosen correctly, because we derived closed form for sums of powers with ease, simply setting parameter for r. For example,

Example 3.2. *For integer $0 \leq t \leq 3$, we have*

$$\Sigma^1 n^3 = 0\left[\tbinom{n+1}{1} - \tbinom{1}{1}\right] + 1\left[\tbinom{n+1}{2} - \tbinom{1}{2}\right] + 6\left[\tbinom{n+1}{3} - \tbinom{1}{3}\right] + 6\left[\tbinom{n+1}{4} - \tbinom{1}{4}\right],$$

$$\Sigma^1 n^3 = 1\left[\tbinom{n}{2} - \tbinom{0}{2}\right] + 7\left[\tbinom{n}{3} - \tbinom{0}{3}\right] + 12\left[\tbinom{n}{4} - \tbinom{0}{4}\right] + 6\left[\tbinom{n}{5} - \tbinom{0}{5}\right],$$

$$\Sigma^1 n^3 = 8\left[\tbinom{n-1}{2} - \tbinom{-1}{2}\right] + 19\left[\tbinom{n-1}{3} - \tbinom{-1}{3}\right] + 18\left[\tbinom{n-1}{4} - \tbinom{-1}{4}\right] + 6\left[\tbinom{n-1}{5} - \tbinom{-1}{5}\right],$$

$$\Sigma^1 n^3 = 27\left[\tbinom{n-2}{2} - \tbinom{-2}{2}\right] + 37\left[\tbinom{n-2}{3} - \tbinom{-2}{3}\right] + 24\left[\tbinom{n-2}{4} - \tbinom{-2}{4}\right] + 6\left[\tbinom{n-2}{5} - \tbinom{-2}{5}\right].$$

From example above, we may observe binomial coefficients of negative upper index, which may looking unfamiliar. However, there is no reason to be afraid of, because negative upper index binomial coefficients are completely deterministic, because $\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$. By setting $t = 0$ into Proposition (3.1) yields one more well-known identity for sums of powers, mentioned in [5, p. 190], and in [7]

Corollary 3.3. *For non-negative integers n, m*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \binom{n+1}{j+1} \Delta^j 0^m.$$

For example,

Example 3.4. *For non-negative integer n, we have*

$$\Sigma^1 n^1 = 1\tbinom{n+1}{1},$$

$$\Sigma^1 n^1 = 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2},$$

$$\Sigma^1 n^2 = 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 2\tbinom{n+1}{3},$$

$$\Sigma^1 n^3 = 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 6\tbinom{n+1}{3} + 6\tbinom{n+1}{4},$$

$$\Sigma^1 n^4 = 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 14\tbinom{n+1}{3} + 36\tbinom{n+1}{4} + 24\tbinom{n+1}{5},$$

$$\Sigma^1 n^5 = 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 30\tbinom{n+1}{3} + 150\tbinom{n+1}{4} + 240\tbinom{n+1}{5} + 120\tbinom{n+1}{6}.$$

From the example above, we observe the coefficients $0, 1, 2, 0, 1, 6, 6, 0, 1, 14, 36, 24, \ldots,$ are defined by Stirling numbers of the second kind $a_k = k!\left\{{n \atop k}\right\}$, which is the sequence A131689 in the OEIS [8]. This leads us to the notion, that Stirling numbers of the second kind $\left\{{n \atop k}\right\}$ recover naturally from forward Newton's interpolation formula for powers. By setting $t = 1$ into Proposition (3.1), we get another well-known special case

Corollary 3.5. *For non-negative integers n, m,*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \binom{n}{j+1} \Delta^j 1^m,$$

mentioned, for instance, in [9]. For example,

Example 3.6. *For non-negative integer n, we have*

$$\Sigma^1 n^0 = 1\tbinom{n}{1},$$

$$\Sigma^1 n^1 = 1\tbinom{n}{1} + 1\tbinom{n}{2},$$

$$\Sigma^1 n^2 = 1\tbinom{n}{1} + 3\tbinom{n}{2} + 2\tbinom{n}{3},$$

$$\Sigma^1 n^3 = 1\tbinom{n}{1} + 7\tbinom{n}{2} + 12\tbinom{n}{3} + 6\tbinom{n}{4},$$

$$\Sigma^1 n^4 = 1\tbinom{n}{1} + 15\tbinom{n}{2} + 50\tbinom{n}{3} + 60\tbinom{n}{4} + 24\tbinom{n}{5},$$

$$\Sigma^1 n^5 = 1\tbinom{n}{1} + 31\tbinom{n}{2} + 180\tbinom{n}{3} + 390\tbinom{n}{4} + 360\tbinom{n}{5} + 120\tbinom{n}{6}.$$

The coefficients $1, 1, 1, 1, 3, 2, 1, 7, 12, 6, 1, 15, \ldots$ is the sequence A028246 in the OEIS [8]. It is interesting to note that the work [9] provide formulas for sums of powers in terms of generalized Stirling numbers of the second kind $\left\{ {k \atop j} \right\}_r$, that is

Proposition 3.7 (Cereceda (2022)). *For non-negative integers n, k*

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1-r}{j+1} + (-1)^j \binom{r+j-1}{j+1} \right] \left\{ {k \atop j} \right\}_r, \tag{1}$$

And

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1+r}{j+1} + \binom{r+1}{j+1} \right] \left\{ {k \atop j} \right\}_{-r}. \tag{2}$$

Formula (1) is a special case of Proposition (3.1) with setting $t \to r$, and upper negated binomial coefficient $\binom{1-t}{k+1}$, by using upper negation formula from Lemma (1.8). We may observe that forward finite difference of powers $\Delta^k t^m$ can be expressed in terms of generalized Stirling numbers of the second kind, that is

$$\Delta^k t^m = k! \left\{ {m \atop k} \right\}_t. \tag{3}$$

This leads us to the observation, that generalized Stirling numbers of the second kind $\left\{ {n \atop k} \right\}_t$ recover naturally from forward Newton's interpolation formula for powers, evaluated in arbitrary integer t. Now, reviewing Formula (2), we notice that it is a special case of Proposition (3.1) with setting $t \to -r$, and by using Equation (3). For example, by setting $t = 4$ into Proposition (3.1) yields rather unusual formulas for sums of powers

Example 3.8. *For non-negative integer n, we have*

$$\Sigma^1 n^0 = 1 \left(\binom{n-3}{1} + \binom{3}{1} \right),$$

$$\Sigma^1 n^1 = 4 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 1 \left(\binom{n-3}{2} - \binom{4}{2} \right),$$

$$\Sigma^1 n^2 = 16 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 9 \left(\binom{n-3}{2} - \binom{4}{2} \right) + 2 \left(\binom{n-2}{3} + \binom{5}{3} \right),$$

$$\Sigma^1 n^3 = 64 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 61 \left(\binom{n-3}{2} - \binom{4}{2} \right) + 30 \left(\binom{n-3}{3} + \binom{5}{3} \right) + 6 \left(\binom{n-3}{4} - \binom{6}{4} \right).$$

The coefficients $1, 4, 1, 16, 9, \ldots$ is the sequence A391633 in the OEIS [8]. As we have all required identities in our disposal, consider the case of double sums of powers. Surely, we derive its formula with ease, and gracefully, by setting $r = 1$ into hockey-stick identity from Lemma (1.6), with additional correction terms $\Sigma^r n^0$. Therefore,

Proposition 3.9 (Double sums of powers)**.** *For non-negative integers n, m, and an arbitrary integer t*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+2}{k+2} - \binom{1-t}{k+1} \Sigma^1 n^0 - \binom{2-t}{k+2} \Sigma^0 n^0 \right] \Delta^k t^m.$$

Note that $\Sigma^1 n^0 = n$, and $\Sigma^0 n^0 = 1$. By applying upper binomial negation formula from Lemma (1.8) yields

Proposition 3.10 (Double sums of powers negated)**.** *For non-negative integers n, m, and an arbitrary integer t*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+2}{k+2} + (-1)^k \binom{k+t-1}{k+1} n + (-1)^{k+1} \binom{k+t-1}{k+2} \right] \Delta^j t^m.$$

For example, by setting $t = 5$ into Proposition (3.10) yields

Example 3.11. *For non-negative integer n, we have*

$$\Sigma^2 n^0 = 1 \left(\tbinom{n-3}{2} + \tbinom{4}{1} n - \tbinom{4}{2} \right),$$

$$\Sigma^2 n^1 = 5 \left(\tbinom{n-3}{2} + \tbinom{4}{1} n - \tbinom{4}{2} \right) + 1 \left(\tbinom{n-3}{3} - \tbinom{5}{2} n + \tbinom{5}{3} \right),$$

$$\Sigma^2 n^2 = 25 \left(\tbinom{n-3}{2} + \tbinom{4}{1} n - \tbinom{4}{2} \right) + 11 \left(\tbinom{n-3}{3} - \tbinom{5}{2} n + \tbinom{5}{3} \right) + 2 \left(\tbinom{n-3}{4} + \tbinom{6}{3} n - \tbinom{6}{4} \right),$$

$$\Sigma^2 n^3 = 125 \left(\tbinom{n-3}{2} + \tbinom{4}{1} n - \tbinom{4}{2} \right) + 91 \left(\tbinom{n-3}{3} - \tbinom{5}{2} n + \tbinom{5}{3} \right) + 36 \left(\tbinom{n-3}{4} + \tbinom{6}{3} n - \tbinom{6}{4} \right)$$
$$+ 6 \left(\tbinom{n-3}{5} - \tbinom{7}{3} n + \tbinom{7}{4} \right).$$

The coefficients $1, 5, 1, 25, 11, 2, \ldots$ is the sequence A391635 in the OEIS [8]. Therefore, generalized formula for r-fold sums of powers is straightforward, repeating the same procedure as we have done for $\Sigma^1 n^m$, and $\Sigma^2 n^m$

Theorem 3.12 (Multifold sums of powers)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s\, n^0 \right] \Delta^k t^m.$$

Which is quite remarkable in its pure binomial form, by using Lemma (1.7)

Proposition 3.13 (Multifold binomial sums of powers)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k t^m.$$

Explicitly,

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n-t+r}{k+r} - \tbinom{r-t}{k+r}\tbinom{n-1}{0} - \tbinom{r-1-t}{k+r-1}\tbinom{n}{1} - \cdots - \tbinom{2-t}{k+2}\tbinom{r+n-3}{r-2} - \tbinom{1-t}{k+1}\tbinom{r+n-2}{r-1} \right] \Delta^k t^m.$$

For example,

Example 3.14. *For non-negative integers r, n, m we have*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+r-0}{k+r} - \sum_{s=0}^{r-1} \tbinom{r-s}{k+r-s}\tbinom{s+n-1}{s} \right] \Delta^k 0^m,$$

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+r-1}{k+r} - \sum_{s=0}^{r-1} \tbinom{r-s-1}{k+r-s}\tbinom{s+n-1}{s} \right] \Delta^k 1^m,$$

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+r-2}{k+r} - \sum_{s=0}^{r-1} \tbinom{r-s-2}{k+r-s}\tbinom{s+n-1}{s} \right] \Delta^k 2^m,$$

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\tbinom{n+r-3}{k+r} - \sum_{s=0}^{r-1} \tbinom{r-s-3}{k+r-s}\tbinom{s+n-1}{s} \right] \Delta^k 3^m.$$

4. Stirling form

We have

Lemma 4.1 (Stirling finite differences)**.** *For non-negative integers k, m, and an arbitrary integer t,*

$$\Delta^k t^m = \sum_{s=0}^{m} \left\{ {m \atop k+s} \right\} \binom{t}{s} (k+s)!$$

Proof. By well-known identity $t^m = \sum_{s=0}^{m} {m \brace s} \binom{t}{s} s!$ □

Thus, from Theorem (3.12) follows

Proposition 4.2 (Multifold Stirling sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \sum_{s=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s\, n^0 \right] {m \brace k+s} \binom{t}{s} (k+s)!$$

By Lemma (1.7), the pure binomial form follows

Proposition 4.3 (Multifold Stirling-Binomial sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \sum_{s=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] {m \brace k+s} \binom{t}{s} (k+s)!$$

5. Eulerian form

We have

Lemma 5.1 (Worpitzky identity)**.** *For non-negative integers t, m*

$$t^m = \sum_{k=0}^{m} \binom{t+k}{m} \left\langle {m \atop k} \right\rangle,$$

where $\left\langle {n \atop k} \right\rangle$ are Eulerian numbers.

Proof. See the (1883) work of J. Worpitzky [10]. □

Thus, we obtain formula for finite differences of powers, in terms of Eulerian numbers $\left\langle {m \atop s} \right\rangle$

Lemma 5.2 (Eulerian finite differences)**.** *For non-negative integers t, k, and an arbitrary integer t,*

$$\Delta^k t^m = \sum_{s=0}^{m} \binom{t+s}{m-k} \left\langle {m \atop s} \right\rangle.$$

Proof. By binomial recurrence $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$, and by Worpitzky identity (5.1). □

Thus, from Theorem (3.12) follows

Proposition 5.3 (Multifold Eulerian sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\sum_{s=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\Sigma^s n^0\right]\binom{t+s}{m-k}\left\langle {m \atop s} \right\rangle.$$

By Lemma (1.7), the pure binomial form follows

Proposition 5.4 (Multifold Eulerian-Binomial sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\sum_{s=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\binom{s+n-1}{s}\right]\binom{t+s}{m-k}\left\langle {m \atop s} \right\rangle.$$

6. Backward difference form

We have

Lemma 6.1 (Forward-Backward relation)**.** *For non-negative integers k, m, and an arbitrary integer t*

$$\Delta^k t^m = \nabla^k (t+k)^m.$$

Thus, from Theorem (3.12) follows

Proposition 6.2 (Multifold Backward sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\Sigma^s n^0\right]\nabla^k (t+k)^m.$$

By Lemma (1.7), the pure binomial form follows

Proposition 6.3 (Multifold Backward-Binomial sums)**.** *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\binom{s+n-1}{s}\right]\nabla^k (t+k)^m.$$

7. Central difference form

We have

Lemma 7.1 (Forward-Central relation). *For non-negative integers k, m, and an arbitrary integer t*

$$\Delta^k t^m = \delta^k \left(t + \frac{k}{2}\right)^m .$$

Thus, from Theorem (3.12) follows

Proposition 7.2 (Multifold Central sums). *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s\, n^0 \right] \delta^k \left(t + \frac{k}{2}\right)^m .$$

By Lemma (1.7), the pure binomial form follows

Proposition 7.3 (Multifold Central-Binomial sums). *For non-negative integers r, n, m, and an arbitrary integer t*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \delta^k \left(t + \frac{k}{2}\right)^m .$$

8. Conclusions

Acknowledgements

References

[1] Knuth, Donald E. Johann Faulhaber and sums of powers. *Mathematics of Computation*, 61(203):277–294, 1993. https://arxiv.org/abs/math/9207222.

[2] J. F. Steffensen. On the definition of the central factorial. *Journal of the Institute of Actuaries (1886-1994)*, 64(2):165–168, 1933. https://www.jstor.org/stable/41137516.

[3] Newton, Isaac and Chittenden, N.W. *Newton's Principia: the mathematical principles of natural philosophy*. New-York, D. Adee, 1850. https://archive.org/details/bub_gb_KaAIAAAAIAAJ/page/466/mode/2up.

[4] Steffensen, Johan Frederik. *Interpolation*. Williams & Wilkins, 1927. https://www.amazon.com/-/de/Interpolation-Second-Dover-Books-Mathematics-ebook/dp/B00GHQVON8.

[5] Graham, Ronald L. and Knuth, Donald E. and Patashnik, Oren. *Concrete mathematics: A foundation for computer science (second edition)*. Addison-Wesley Publishing Company, Inc., 1994. https://archive.org/details/concrete-mathematics.

[6] Donald E. Knuth. Two notes on notation, 1992. https://arxiv.org/abs/math/9205211.

[7] Thomas J. Pfaff. Deriving a formula for sums of powers of integers. *Pi Mu Epsilon Journal*, 12(7):425–430, 2007. https://www.jstor.org/stable/24340705.

[8] Sloane, Neil J.A. and others. The on-line encyclopedia of integer sequences. https://oeis.org/, 2003. https://oeis.org/.

[9] Cereceda, José L. Sums of powers of integers and generalized Stirling numbers of the second kind. *arXiv preprint arXiv:2211.11648*, 2022. https://arxiv.org/abs/2211.11648.

[10] J. Worpitzky. Studien über die bernoullischen und eulerschen zahlen. *Journal für die reine und angewandte Mathematik*, 94:203–232, 1883.

Mathematica programs

Use this GitHub repository to validate the results using Mathematica programs.

Mathematica Function	**Validates / Prints**
`ValidateMultifoldSumsOfPowers.txt`	Validates Thm. (3.12)
`ValidateMultifoldBinomialSumsOfPowers.txt`	Validates Prop. (3.13)

Metadata

- **Initial release date:** December 24, 2025.
- **Current release date:** June 27, 2026.
- **Version:** `Local-0.1.0`
- **MSC2010:** 05A19, 05A10, 41A15, 11B83
- **Keywords:** Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Newton's series, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Combinatorics, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, OEIS

- **DOI:** https://doi.org/10.5281/zenodo.18040979
- **Web Version:** kolosovpetro.github.io/sums-of-powers-newtons-formula
- **Sources:** github.com/kolosovpetro/NewtonsFormulaAndSumsOfPowers

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